

ATTORNEY SHIELD

ASI 2.0 - Native Android & iOS

ASI 2.0 Development Plan

Native Android & iOS member apps

Draft for approval - 2026-08-10

Colour system: Attorney-Shield Color Scheme Review

Repository: samson-phillip/asi-claude

DRAFT FOR APPROVAL

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Date: 2026-08-10 **Scope:** Native Android (kotlin) and iOS (swift) member apps **Status:** Draft for approval - no code written yet

1. Objective

Ship native Android and iOS member apps that match member-client's behaviour and API usage, rendered in the Attorney Shield visual system from the colour PDF and the CodePen onboarding reference.

2. Reference authority

Rank	Source	Governs
1	design/color-system.md (from the colour PDF)	All colour values
2	CodePen "Attorney - Shield App V6"	Layout, typography, screen inventory, flow
3	member-client	Behaviour, API contracts, state machines, errors

Full rationale and the resolved PDF-vs-CodePen conflicts are in [design/color-system.md](#). Two rules that will otherwise bite: member-client's **blue/violet CSS is not the palette**, and **never white text on gold** (3.13:1, fails WCAG AA).

3. Scope reality check - read this before estimating

The design board is **far larger than** member-client, and the API covers only a fraction of it. Figures below are counted from the reference itself ([notes/design-reference-codepen.md](#)), not from its masthead.

The CodePen board actually contains 66 phone frames - 7 welcome-carousel frames plus 59 numbered/annotated screens. Of the 66, **9 carry web browser chrome** (04, 05, 06, 07, V1, T1-T4) and **57 are native**. Its masthead claim of "35 screens · 5 stages" is wrong on both counts: 35 is merely the highest plain screen number, and only Stages **1, 4 and 5** exist - there is no Stage 2 or Stage 3 header anywhere in the document.

member-client **implements 3 screens** - LoginScreen -> HomeScreen -> CallScreen.

Payment is on the web - this is the key structural finding

Plan choice, Stripe checkout and confirmation are **web pages on attorney-shield.com**, not native screens. A deep link then hands the member back to the app with their email pre-filled, and native registration resumes at screen 08. So payment always precedes phone verification.

```
native welcome (1-6) -> WEB 04 plan -> 05 Stripe -> 06 confirm -> 07 deep link
-> native 08 phone -> 09 verify -> 10 details -> 11 address -> 12 PIN -> home
```

Those web frames sit inside the Stage 1 section and therefore **inherit a "Native App" badge despite being browser frames** - a labelling bug in the reference, not an instruction to build them natively. (A "Web" badge style is defined in its stylesheet and never applied to anything.)

This materially reduces native scope: **we do not build checkout**. It also means the real prices are **FreedomPlus** \$16/mo and **FreedomFAMILY** \$38/mo, plus a \$0-today 7-day trial - *not* the \$19/\$179/\$29 figures on the colour PDF's marketing-site preview, which are a different surface.

What is still blocked

Payment being on web does **not** clear the blockage - it moves it. Every documented operation is:

- GraphQL POST /query - login, user, casesByUser, adminIncidentTypeList + adminLanguageList, partnerAttorneys
- REST POST {API_BASE_URL}/api/vonage/video/member-call

There is **no endpoint** behind any of the following native screens:

Native feature	Screens	Endpoint
Phone entry + SMS/OTP verification	08, 09	none
Personal details, address	10, 11	none
4-digit PIN set / verify / end session	12, 34	none
Document vault ("Digital Glovebox")	14, 14A-14D, 31	none
Situation preferences (pick 3)	13B, 13C, 27B	none
Activity timeline	32	none
Plan / payment method / family sub-accounts	33A, 33B, 33D	none
Notifications & nudge system	15, 22-26	none
Trial gate + in-app conversion	V2, T5-T8	none
Guest session & feature gates	G1-G3	none

CLAUDE.md forbids inventing endpoints. Therefore:

Native registration (08-12) and the entire post-registration feature set are BLOCKED on backend work that does not exist yet. They can be built as UI against mocks, but not as functioning features.

What *is* unblocked is the spine: welcome -> login -> home -> connect -> call. That is Phases 0-2. Everything above is Phase 3+, and needs §9 answered first.

What we can build today

Capability	API support	Buildable now
Email/password login	login	Yes
Member profile / org	user	Yes
Case -> jurisdiction/partner/case resolution	casesByUser	Yes
Incident-type tiles (multilingual)	adminIncidentTypeList + adminLanguageList	Yes
Attorney pre-selection	partnerAttorneys	Yes
Place video call + Vonage credentials	member-call	Yes
Welcome/onboarding carousel (7 frames)	none needed (static)	Yes
Deep-link return from web checkout	none needed (client-side)	Yes
Plan choice / checkout / confirmation	not ours - web	Out of native scope
Native registration 08-12, vault, family, nudges	none	No Blocked (§3)
Trial gate, guest gate	none	No Blocked (§3)
Token refresh	none used	No See §8

4. Target architecture

Mirror structure across platforms so one review covers both.

```
<repo>/
core/design/ tokens (colour, type, spacing), reusable components
core/network/ GraphQL client, REST client, error mapping
core/session/ auth + resolved member context, persistence
feature/auth/ login
feature/home/ greeting, attorney chips, incident tiles
feature/call/ Vonage video, phase machine
```

Android (kotlin)

- Jetpack Compose, Material 3 with a fully overridden colour scheme
- MVVM, StateFlow for state; Hilt for DI
- Ktor or OkHttp + kotlinx.serialization. **Raw** POST /query, not Apollo - faithfulness to api.ts's hand-written queries matters more than codegen, and it avoids requiring a schema we do not have
- Vonage Video Android SDK
- EncryptedSharedPreferences for the session

iOS (swift)

- SwiftUI, @Observable view models
- URLSession + Codable, same raw-GraphQL approach
- Vonage Video iOS SDK
- Keychain for the session

Design system as code

Ship design/color-system.md §6 tokens as a single source file per platform (AsiColors.kt, AsiColors.swift). **No raw hex literals in UI code** - this is a lint/review gate, not a convention (§ testing plan).

5. Behaviour to port verbatim

From member-client. These are the details that make the app correct:

- 1 **Call routing** - send attorneyId when the member pre-selected one, **else** queueId. Never both.
- 2 **Call phase machine** - exactly starting -> connecting -> live -> ended, plus no-attorney and error terminals.
- 3 409 **on** member-call = NoAttorneyError, a *retryable* message. Never a crash, never a generic failure.
- 4 **Incident labels are multilingual** - English -> org default language -> first available -> humanized code. Sort by sortOrder.
- 5 **Location is best-effort** - never block a call on geolocation. Resolve null on unsupported/denied/timeout.
- 6 listAttorneys **never throws** - empty list means auto-match/queue only, and the chip row hides entirely.
- 7 **Login never blocks on reference lookups** - user and casesByUser failures fall back to DEV_DEFAULTS.
- 8 **Validate video credentials** - reject a member-call response missing apiKey/sessionId/token/callId/videoRoomId.
- 9 **Publisher/session teardown** on unmount - unpublish, destroy, disconnect, each independently guarded.

6. Phases

Each phase ends with: unit tests green, emulator + simulator verified against member-client in Claude Browser, and a journal entry.

Phase 0 - Foundations (*no product surface*) Scaffold both projects with test harnesses. Encode design tokens and verify the contrast table programmatically. Establish CI-runnable test commands. *Exit*: ./gradlew test and xcodebuild test both run green on a trivial test; token contrast test passes.

Phase 1 - Network + session GraphQL and REST clients, error mapping, session persistence, member-context resolution with DEV_DEFAULTS fallback. *Exit*: login against dev returns a token and resolves org/jurisdiction; unit tests cover the fallback paths in §5.7.

Phase 2 - Welcome -> Login -> Home -> Call (the working spine) The three `member-client` screens in the Attorney Shield visual system, plus the 7-frame welcome carousel (static, no API needed) and the **deep-link return handler** (Universal Links / App Links with email pre-fill and the "Open app manually" fallback) - registration cannot continue without it. *Exit*: a real video call connects on both platforms; all six call phases reachable in tests; deep link resumes the app from a cold start; behaviour parity checklist (§5) signed off.

Phase 3 - Registration and member features (*gated* - see §3) Screens 08-12 and the post-registration set (vault, family, activity, nudges, trial/guest gates). **Requires §9.1 and §9.2 answered first.** Built against mocks only if we explicitly choose the mock path.

Phase 4 - Hardening Accessibility audit, dynamic type, rotation, offline/airplane-mode, token expiry, permission-denied paths, performance.

Phases 0-2 are the committed plan. Phase 3 is deliberately unestimated until §9 is answered.

7. Cross-cutting requirements

- **Accessibility:** WCAG AA (4.5:1 text / 3:1 UI) enforced by test, not review. Respect OS font scaling and screen readers; every icon-only control labelled.
- **Permissions:** camera and microphone are required for the call; location is optional and must degrade silently. Request at point of use with a clear rationale, and handle permanent denial.
- **Deep links** are a product requirement, not plumbing: iOS Universal Links and Android App Links from attorney-shield.com, resuming registration with the email pre-filled, plus the manual fallback. Must survive a cold start.
- **No secrets in the repo.** `apiKey` arrives per-call from the backend - never bundle Vonage credentials.
- **Config per environment**, mirroring `config.ts` (`GRAPHQL_URL`, `API_BASE_URL`, `DEV_DEFAULTS`, `CALL_TTL_SECONDS = 3600`).

8. Risks

Risk	Impact	Mitigation
App Store / Play compliance on web checkout. The flow sells a digital subscription through web Stripe and deep-links back. Apple's rules on purchases for digital content and on steering users to external payment are the single most likely cause of a rejection here, and Google has parallel constraints	Could block release outright - not just delay it	Settle the store strategy before Phase 3 , not at submission. The reference already anticipates part of this (33A carries App Store's required delete-account wording; no in-app plan changes). Needs a definitive answer on external-purchase entitlements vs in-app purchase
Missing endpoints for native registration, vault, family, nudges, trial/guest gates	Blocks everything past the call spine (§3)	Resolve §9.1 before Phase 3; Phases 0-2 are unaffected

Risk	Impact	Mitigation
Deep-link return is load-bearing. Registration cannot continue without it, and the reference plans to branch it by origin (app-initiated vs website-mobile)	A broken link strands members mid-signup, after payment	Treat as a Phase 2 deliverable with its own tests, not an afterthought. Universal Links + App Links, with the "Open app manually" fallback
No token refresh exists - login returns a refreshToken but api.ts never sends it; there is no refresh operation	Sessions die with no recovery path; native apps are backgrounded far longer than a web tab, so this hurts more than it does on web	Spike in Phase 1. Interim: detect auth failure and re-prompt login cleanly
Web->native video SDK gap - reference uses @vonage/client-sdk-video; native needs the Vonage Video Android/iOS SDKs	Could invalidate Phase 2 estimates	De-risk in Phase 0 with a spike proving apiKey/sessionId/token connect natively
member-call has no auth today	Abuse surface	Flag to backend; do not design around it
CodePen is v11 / "locked reference" but PDF conflicts with it	Rework	Authority order in §2; conflicts already resolved
409 handling regressions	Members see a crash at the worst moment	Explicit test case, both platforms
Reference app not yet run locally	Parity checks can't start	Stand it up in Phase 0 (needs serve-proxy.mjs)

9. Open questions - need answers before Phase 3

- Endpoints for native registration and the post-registration feature set - do they exist?** *Partially answered by the reference:* checkout is web, so payment is out of native scope. But screens 08-12 (phone, OTP, details, address, PIN) and the vault / family / activity / nudge / trial / guest features have no endpoints at all (§3). Is the intent that backend builds them, or that we ship UI against mocks? **Still the highest-value question.**
- App Store / Play strategy for a web-purchased subscription** - external purchase entitlement, or in-app purchase? This decides whether store review is on the critical path, and it is cheaper to answer now than at submission.
- Token refresh** - is there a refresh operation on the gateway we should use?
- Error/danger colour** - the palette forbids #FF4500 and supplies no error colour. What should errors and hang-up use? (See colour-system §6.)
- Which CodePen palette values are stale** - confirm the PDF supersedes Deep Navy #0A1626, Live Green #2E9E5B, Trust Steel #1A5FA8.

- 6 **Family plan capacity is stated four different ways** in the reference - "covers up to 5" / "includes 3, add up to 2 more" / "You + 3" / "You + up to 4" / "3 of 5 on your plan". Whether 5 includes the primary account is genuinely unresolved. **Must be settled before building the member stepper.**
- 7 **Frame geometry** - the board is drawn at 292x600 px but labelled "iPhone · 375pt". Confirm 375pt is the intent, in which case all reference measurements need rescaling 1.284x.
- 8 **Guest user type** - the reference says guests browse the real home and hit a gate on every member feature. Confirm nothing privileged is reachable.

10. Immediate next steps

- 1 Get §9.1 and §9.3 answered.
- 2 Phase 0: scaffold both repos, encode tokens, **run the Vonage native spike.**
- 3 Stand up member-client locally for parity comparison.